SCHRÖDINGER EVOLUTION FROM THE BARE CAUSAL ACCUMULATION LAW AT $\alpha \to 1^-$: FEEDBACK ON THE ACCUMULATED PAST, UNITARITY IN THE LIMIT, AND CARRIER ENTANGLEMENT AS TIME DILATION

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Abstract. We read the Causal Accumulation Law (CAL) as a transducer: each output state becomes the next current. This is a feedback loop of repeated impulses on the accumulated past. Nothing else is added. There is no phase prefactor on the memory, no coupling constant, no time unit, and no imposed field equation $T = 0$. Time is counted in updates and quoted in radians of the carrier's own rotation.

- *Rotation from negative feedback.* Below $\alpha = 1$ the memory term enters with the sign $-(1-\alpha) < 0$, so the accumulated past pulls the current back. The current rotates by $\sqrt{1-\alpha}$ per update. The i of the Schrödinger equation is this rotation: the phase is a property of the source current, not a factor in the law.
- *Schrödinger's dispersion and the mass.* The rotation slows with wavenumber as $\sqrt{\hat{g}(k)}$, where $\hat{g}$ is the kernel's spatial profile. Per radian of rest rotation the kinetic phase is $\langle r^2 \rangle k^2/12$, so $\hbar/2m = \langle r^2 \rangle/12$, set by the kernel's spatial width alone. The regime is long wavelengths, and the leading correction is of order k^4.
- *Unitarity in the limit.* For $\alpha < 1$ the evolution is a contraction, with two losses. There is a uniform loss of $(\pi/4)(1-\alpha)$ per radian, and a diffusive admixture of relative size $\sqrt{1-\alpha}$. Each mode evolves as $\exp[-(i+\sqrt{1-\alpha})(\hbar/2m)k^2 t]$. Both losses vanish as $\alpha \to 1^-$, where the evolution is unitary. Unitarity is derived, not posited, but it is exact only in the limit, and the size of the departure below it is stated.
- *Orientation.* The direction of rotation is inherited from the source's past. A current that has always rotated one way keeps doing so, with leakage of order 10^{-8}. A zero past excites both directions, and then the Born readout drifts. Born needs one orientation.
- *Feedback across $\alpha = 1$.*
  - At $\alpha = 1$ the past decouples and the carrier freezes.
  - Above 1 the feedback is positive. The causal response runs away at about $\sqrt{\alpha - 1}$ per update, and a bounded response must precede its source. These are the two horns of the Abraham–Lorentz–Dirac problem.
  - Braking together with a fading past can settle the growth. Braking alone cannot, which is integral windup.
- *Memory, loss and Markov structure.*
  - A normalizable memory has no undamped mode. The power law escapes because its total weight is infinite.
  - The edge is Markov through the fundamental theorem of calculus.
  - Information loss is contraction from rank 2 to rank 1. It is absent in the free edge and present in interaction and readout.

The structural results are kept. The edge is Galilean and carries no interaction. One carrier $J \in M_2(\mathbb{C})$ is two factors, whose concurrence is $1/\gamma$ of its current. The trace readout survives tensor composition.

Numerical runs of the iterated law confirm each statement.

ORCID: 0009-0000-0133-2291. Companion to *The Causal Accumulation Law: The Four Corners of α, Algebraic Closure, and the Standard Model* [1] and to *The $\alpha = 1$ Unitary Skeleton and the Emergence of Schrödinger Evolution in the δ-Kernel Limit of the CAL* [2], whose unitarity posit it replaces. This version supersedes an earlier one whose field equation carried a phase prefactor, a coupling constant and a fixed time unit that are not in the CAL. Dated November 13, 2026.

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1. Scope and claim

The companion paper [2] obtains the Schrödinger form at the edge of the (α, kernel) table adjoining $\alpha = 1$ from below. It consumes an input: Posit 1 assumes a unitary flow and then defines the generator from it. An earlier version of the present paper replaced that posit with a field equation, $T = 0$. To make $T = 0$ produce the Schrödinger equation, it added three things that the CAL does not contain:

- a phase prefactor on the memory;
- a coupling constant;
- a fixed time unit, scaled together with α.

This version removes all of them. It uses the CAL as written, read as a transducer.

> *Iterate the CAL: each output state becomes the next current (Axiom 3). For $\alpha < 1$ the accumulated past pulls back, and the current rotates by $\sqrt{1-\alpha}$ per update (Theorem 5.1). The rotation slows with wavenumber as Schrödinger's dispersion does, with $\hbar/2m = \langle r^2 \rangle/12$ in units of the rest rotation (Corollary 5.2). The evolution is a contraction that becomes unitary as $\alpha \to 1^-$ (Theorem 5.3). The direction of rotation is inherited from the source's past (Proposition 5.4). At $\alpha = 1$ the carrier freezes, and above it the feedback runs away or must anticipate its source (Theorem 5.5). Normalizable memory cannot sustain a rotation (Theorem 6.2). The structural results on causality, interaction and readout are kept (§7).*

Remark 1.1 (Ledger). *Inputs:*

- the CAL (Axiom 1);
- the free, flat sector with a power-law memory (Definitions 2.2–2.3);
- the phase as a property of the source (Axiom 2);
- the update rule (Axiom 3);
- a one-way past (Axiom 4).

Derived: §§4–7.

Not used: a prefactor on the memory, a coupling constant, a time unit, any double scaling, or $T = 0$ as a law.

Not claimed:

- a value for the mass in external units;
- interaction between carriers;
- Lorentz structure of the edge's dynamics;
- a rigorous remainder estimate for the asymptotic expansion of Theorem 5.1, which is formal and confirmed numerically.

2. Definitions

Definition 2.1 (Causal Weyl integral)**.** For bounded f and $\alpha \in (0, 1)$, the causal (left-sided) Weyl integral is

$$W^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \int_0^\infty s^{\alpha-1} f(t-s)\, ds.$$

It integrates over the past only. On an oscillation e^{-iEt} its symbol is $(-iE)^{-\alpha} = |E|^{-\alpha} e^{i\,\mathrm{sgn}(E)\,\alpha\pi/2}$. An order-$\alpha$ accumulation turns an oscillation by α quarter turns.

Definition 2.2 (Memory in update units)**.** Time advances in updates $n \in \mathbb{Z}$. The order-α power-law memory assigns to the lag $j \geq 1$ the weight

$$w_j = \frac{j^\alpha - (j-1)^\alpha}{\Gamma(\alpha+1)},$$

which is the integral of $\tau^{\alpha-1}/\Gamma(\alpha)$ over $[j-1, j]$. Its generating function is

$$W(z) = \sum_{j \geq 1} w_j z^{-j}.$$

The spatial profile is a normalized isotropic $g \geq 0$ with Fourier transform $\hat{g}(k) = 1 - \langle r^2 \rangle k^2/6 + O(k^4)$. The numerical runs use $\hat{g} = e^{-\langle r^2 \rangle k^2/6}$.

Definition 2.3 (Carrier and sector)**.** The carrier is $J \in \mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$. In the free, flat sector the bitensor coupling, the parallel transport and the internal fiber transport of the CAL are the identity, $\Pi P \Phi = \mathbf{1}$. The law then acts on each spatial Fourier mode $e^{ik\cdot x}$ separately, and on each matrix entry of J in the same way.

Definition 2.4 (Relational time)**.** No time unit is used. If the carrier rotates by φ_0 per update at $k = 0$ (Theorem 5.1), then times are quoted in radians of that rest rotation, $t = \varphi_0\, n$. Rates are dimensionless ratios of rates.

3. Axioms

Axiom 1 (Causal accumulation law; C8 of [1])**.**

$$T(x) = \alpha J(x) - (1-\alpha) \int_{J^-(x)} k(\tau; x, x')\, \Pi(x, x') P(x, x') \Phi(x, x')\, J(x')\, d^4x'.$$

The kernel and the transports are real.

Axiom 2 (The phase belongs to the source)**.** The channel angle $\theta = \alpha\pi/2$, i.e. $\alpha = 2\theta/\pi$, is what an order-α accumulation does to the source's phase channel $e^{iS/\hbar}$ (Definition 2.1). The real channel $e^{-H/\tau}$ is not turned. No phase factor multiplies the memory term.

Axiom 3 (Update rule)**.** Each output state becomes the next current: $J_{n+1} = T_n$.

T and J are both rank-1 currents, so the rule is type-consistent. It is a feedback loop: the carrier's accumulated past acts on it at every update. This is the edge's dynamics. It is not used at $\alpha = 2$, where the term $2J$ alone would double the current at every update; there the balance reading of [4] applies.

Axiom 4 (Orientation)**.** The current's past has rotated one way: $J_n = c\, e^{i\varphi_h n}$ for all $n < 0$, with φ_h real. This is the role that the orientation σ played in [2], now as a property of the source rather than a factor.

With Definitions 2.2–2.3, Axioms 1 and 3 give, mode by mode,

$$J_{n+1} = \alpha J_n - (1-\alpha)\,\hat{g}(k) \sum_{j\geq 1} w_j\, J_{n-j}. \tag{1}$$

Below $\alpha = 1$ the accumulated past enters with a negative sign. Above 1 it enters with a positive sign.

4. Lemmas

Lemma 4.1 (Weyl convergence)**.** *Let $\omega \in \mathbb{R} \setminus \{0\}$. The improper integral $\int_0^\infty s^{\alpha-1} e^{i\omega s} ds$ converges if and only if $0 < \alpha < 1$. Its value is then $\Gamma(\alpha)(-i\omega)^{-\alpha}$.*

Proof. This is the standard Dirichlet test. The value follows by rotating the contour onto the imaginary axis [10]. □

Lemma 4.2 (Exact memory sum)**.** *For $0 < \alpha < 1$ and $z = e^{i\varphi}$, with φ real and nonzero modulo 2π,*

$$W(e^{i\varphi}) = \big(1 - e^{-i\varphi}\big) \sum_{k\in\mathbb{Z}} \big(i(\varphi + 2\pi k)\big)^{-\alpha-1}, \tag{2}$$

with principal powers. The right side is analytic in φ off the points $2\pi\mathbb{Z}$. It continues W from $|z| > 1$, where the series converges, across the unit circle. The terms with $k \neq 0$ sum to two Hurwitz zeta functions.

Proof. Write $x = z^{-1}$. Then

$$\sum_{j\geq 1} \big(j^\alpha - (j-1)^\alpha\big) x^j = (1-x) \sum_{j\geq 1} j^\alpha x^j = (1-x)\,\mathrm{Li}_{-\alpha}(x).$$

Jonquière's formula gives $\mathrm{Li}_s(e^\mu) = \Gamma(1-s) \sum_k (2\pi i k - \mu)^{s-1}$ for $\mathrm{Re}\, s < 0$ [8, 9]. With $s = -\alpha$ and $\mu = -i\varphi$, the factor $\Gamma(1+\alpha)$ cancels the one in w_j. Numerically, (2) agrees with direct summation to 10^{-14} wherever the series converges, and with an independent arbitrary-precision evaluation to 3×10^{-16} (§8). □

Lemma 4.3 (Modes)**.** *Let*

$$F(z) = z - \alpha + (1-\alpha)\,\hat{g}\, W(z).$$

1. *A root z_0 of F on $|z| = 1$ gives an exact solution $J_n = z_0^n$ of (1) for all $n \in \mathbb{Z}$, with the memory over the infinite past summed as in (2).*
2. *A root with $|z_0| < 1$, continued through (2), governs a decaying mode z_0^n.*
3. *For a zero past and $J_0 \neq 0$, the z-transform of the solution is $zJ_0/F(z)$. The solution is the sum of residues at the roots of F and a branch-cut contribution from $z = 1$.*
4. *For a one-way past (Axiom 4), that past adds to the memory at update $n \geq 0$ the known term*

$$c\, e^{i\varphi_h n} \Big(W(e^{i\varphi_h}) - \sum_{j=1}^{n} w_j e^{-i\varphi_h j} \Big).$$

Proof. For (1), insert z_0^n into (1): the memory term is $z_0^n W(z_0)$. For (3), multiply (1) by z^{-n-1} and sum over $n \geq 0$. For (4), the part of the sum with $j > n$ runs over the history. □

Lemma 4.4 (Flat weights)**.** *If $w_j \equiv 1$, which is the limit of w_j as $\alpha \to 1$ at every fixed j, then $W(z) = 1/(z-1)$ and*

$$(z-1)F(z) = z^2 - (1+\alpha)z + \alpha + (1-\alpha)\hat{g}.$$

At $k = 0$, where $\hat{g} = 1$:

1. *for $\alpha < 1$ the roots are $e^{\pm i\varphi}$ with $\cos\varphi = (1+\alpha)/2$, exactly on the unit circle;*
2. *for $\alpha = 1$ there is a double root $z = 1$;*
3. *for $\alpha > 1$ the roots are $e^{\pm s}$ with $\cosh s = (1+\alpha)/2$.*

In each case the angle φ, or the exponent s, equals $\sqrt{|1-\alpha|}\,(1 + O(|1-\alpha|))$.

Proof. $\sum_{j\geq 1} z^{-j} = 1/(z-1)$ for $|z| > 1$. The product of the roots is $\alpha + (1-\alpha)\hat{g}$, which equals 1 at $k = 0$, and their sum is $1 + \alpha$. □

5. Schrödinger evolution from feedback

Write $\varepsilon = 1 - \alpha$, and write the co-rotating root as $z = e^{i\varphi}$, with $\operatorname{Re}\varphi > 0$ the rotation per update and $\operatorname{Im}\varphi$ the decay per update.

Theorem 5.1 (Rotation from negative feedback)**.** *For fixed $\hat{g} \in (0, 1]$ and $\varepsilon \to 0^+$, F has a pair of complex-conjugate roots near the unit circle. The co-rotating one is*

$$\varphi = \sqrt{\varepsilon\hat{g}}\,\big(1 + O(\varepsilon \ln \varepsilon)\big) + i\Big[\tfrac{1}{2}\varepsilon(1-\hat{g}) + \tfrac{\pi}{4}\varepsilon^{3/2}\sqrt{\hat{g}}\Big] + O(\varepsilon^2 \ln \varepsilon).$$

So the current rotates by $\sqrt{\varepsilon}$ per update at $k = 0$, and decays slowly.

Proof (formal asymptotics). In (2) with $\alpha = 1 - \varepsilon$ and $\varphi \to 0$:

- the $k = 0$ term gives $(1 - e^{-i\varphi})(i\varphi)^{-2+\varepsilon} = (i\varphi)^{-1+\varepsilon} - \frac{1}{2}(i\varphi)^{\varepsilon} + O(\varphi)$;
- the terms with $k \neq 0$ contribute $O(\varphi)$;
- $(i\varphi)^{\varepsilon} = 1 + \varepsilon \ln(i\varphi) + \ldots$, with $\ln(i\varphi) = \ln\varphi + i\pi/2$.

Hence

$$F = i\varphi - \frac{\varphi^2}{2} + \varepsilon + \frac{\varepsilon\hat{g}}{i\varphi} - \frac{\varepsilon\hat{g}}{2} + \frac{\varepsilon^2\hat{g}\ln(i\varphi)}{i\varphi} + O(\varphi^3, \varepsilon\varphi).$$

Order by order, with $\varphi \sim \sqrt{\varepsilon}$:

- *Order $\varepsilon^{1/2}$.* $i\varphi + \varepsilon\hat{g}/(i\varphi) = 0$, so $\varphi_0 = \sqrt{\varepsilon\hat{g}}$, with $F'(\varphi_0) = 2i$.
- *Order ε.* The constant is $-\varphi_0^2/2 + \varepsilon - \varepsilon\hat{g}/2 = \varepsilon(1-\hat{g})$, so $\varphi_1 = i\varepsilon(1-\hat{g})/2$.
- *Order $\varepsilon^{3/2}$.* The logarithmic term gives $\varphi_2 = \frac{1}{2}\varepsilon^{3/2}\sqrt{\hat{g}}\,(\ln\varphi_0 + i\pi/2)$.

The other terms of order $\varepsilon^{3/2}$ are imaginary in F, so they shift only the real part of φ. Section 8 confirms the expansion down to $\varepsilon = 10^{-7}$. There the measured decay per radian, divided by ε, is 0.78667, against $\pi/4 = 0.78540$. □

The decay comes from the fractional order. With flat weights the root at $k = 0$ lies exactly on the unit circle (Lemma 4.4). The power law turns the past by $(1-\varepsilon)\pi/2$ rather than by a full quarter turn, and the mismatch $\varepsilon\pi/2$, entering through $\ln(i\varphi)$, is the term $(\pi/4)\varepsilon^{3/2}$.

Corollary 5.2 (Dispersion and mass)**.** $\operatorname{Re}\varphi(k)/\operatorname{Re}\varphi(0) = \sqrt{\hat{g}(k)}\,(1 + O(\varepsilon \ln \varepsilon))$. *In rest radians (Definition 2.4), mode k therefore lags the rest rotation by the phase*

$$\big(1 - \sqrt{\hat{g}(k)}\big)\,t = \frac{\langle r^2\rangle}{12}\,k^2 t + O(k^4 t).$$

This is free Schrödinger evolution with

$$\frac{\hbar}{2m} = \frac{\langle r^2 \rangle}{12}$$

in units of the rest rotation. The mass is set by the kernel's spatial width alone. The Schrödinger form holds at long wavelengths, $\langle r^2 \rangle k^2 \ll 1$. *For a Gaussian profile the lag is exactly* $1 - e^{-\langle r^2 \rangle k^2/12}$ *at leading order in* ε.

Proof. Theorem 5.1 gives $\varphi_0(k) = \sqrt{\varepsilon \hat{g}(k)}$. Expand $\sqrt{\hat{g}} = 1 - \langle r^2 \rangle k^2/12 + O(k^4)$. □

On the co-rotating branch $e^{+i\varphi n}$, the lag $e^{-i(\hbar/2m)k^2 t}$ has the sign of a positive mass.

Theorem 5.3 (Unitarity in the limit)**.** *With the rest rotation removed and* t *in rest radians, mode* k *evolves as*

$$\exp\Big[- i\big(1 - \sqrt{\hat{g}}\big)t - \Big(\tfrac{1}{2}\sqrt{\varepsilon}\,(1 - \hat{g}) + \tfrac{\pi}{4}\varepsilon\sqrt{\hat{g}}\Big)t\Big],$$

which at long wavelengths is

$$\exp\Big[- \big(i + \sqrt{1 - \alpha}\big)\frac{\hbar}{2m}k^2 t - \frac{\pi}{4}(1 - \alpha)\,t\Big].$$

(1) *For* $\alpha < 1$ *every mode decays, so the evolution is a contraction.*
(2) *The decay has two parts:*
   - *a uniform loss of* $(\pi/4)(1 - \alpha)$ *per radian;*
   - *a diffusive admixture, whose ratio to the kinetic phase is* $\frac{1}{2}\sqrt{\varepsilon}\,(1 + \sqrt{\hat{g}}) - \frac{\pi}{4}\varepsilon \to \sqrt{1 - \alpha}$.

(3) *As* $\alpha \to 1^-$ *both vanish, and the evolution converges mode by mode to the unitary group* $e^{-i(\hbar/2m)k^2 t}$. *The convergence is exact Schrödinger in the further long-wavelength limit* $\langle r^2 \rangle k^2 \to 0$ *at fixed* $\hbar/2m$.

Proof. Divide Theorem 5.1 by $\operatorname{Re}\varphi(0) = \sqrt{\varepsilon}\,(1 + O(\varepsilon \ln \varepsilon))$, and use $1 - \hat{g} = (1 - \sqrt{\hat{g}})(1 + \sqrt{\hat{g}})$. □

Unitarity is therefore derived, but it is exact only at the edge. Below it, the departure is a small, stated contraction.

The evolution has two parts. The rotation belongs to the source's phase channel $e^{iS/\hbar}$. The diffusive part belongs to its real channel $e^{-H/\tau}$, and that admixture vanishes in the limit.

Proposition 5.4 (Orientation is inherited from the past)**.** (1) *For flat weights at* $k = 0$, *a one-way past* $c\,e^{i\varphi n}$ *with* $\cos\varphi = (1 + \alpha)/2$ *continues as* $c\,e^{i\varphi n}$ *for all* n. *The coefficients of* (1) *are real, so the counter-rotating mode* $e^{-i\varphi n}$ *is never excited.*
(2) *For power-law weights, a one-way past rotating at* $\operatorname{Re}\varphi(k)$ *excites the counter-rotating mode only at a level that vanishes with* ε. *At* $\varepsilon = 10^{-4}$ *the counter-rotating fraction of the norm is* 10^{-7}–10^{-8} *(Table 4).*
(3) *A zero past excites both directions with comparable weight, as a spring started from rest swings both ways.*
(4) *A history truncated after a few cycles fails. The memory is nearly flat, so every past update counts, and a two-cycle history sends* 70–97% *of the norm into the counter-rotating mode. The whole past is required.*

Proof. Part (1) is Lemma 4.3(1) with Lemma 4.4. Parts (2)–(4) are numerical, §8. □

The wave at a point is therefore the carrier's accumulated causal past evaluated there. It is not a spread in space at one instant.

Theorem 5.5 (Feedback across $\alpha = 1$)**.** (1) *At* $\alpha = 1$ *the memory term vanishes. The past decouples, and* (1) *gives* $J_{n+1} = J_n$: *the carrier is frozen.*

(2) *For $\alpha > 1$ the feedback is positive, and F has a real root $z_* > 1$. The causal response therefore grows like z_*^n, at about $\sqrt{\alpha - 1}$ per update. With flat weights the two roots are $e^{\pm s}$, with $\cosh s = (1 + \alpha)/2$.*

(3) *The only bounded response to a source at n_0 decays in both directions from n_0. With flat weights it is proportional to $e^{-s|n-n_0|}$. It is nonzero before the source acts, so it is advanced. Above 1 one must choose between a causal runaway and a bounded preacceleration: the two horns of the Abraham–Lorentz–Dirac problem [6].*

(4) *Add a brake, $J_{n+1} = (1 - b)\, T_n$ with $0 \le b < 1$, and a fading past, with weights $w_j e^{-\lambda j}$. Then settling requires*

$$(1 - b)\big(\alpha + (\alpha - 1)W_\lambda(1)\big) < 1.$$

If the past never fades, $\lambda = 0$, then $W(1) = \infty$ and no brake settles the growth. This is integral windup.

Proof. Part (1) is immediate.

For (2) and (4): for real $z > 1$, $F(z) = z - (1 - b)\alpha - (1 - b)(\alpha - 1)W_\lambda(z)$ is continuous, and $F(z) \to +\infty$ as $z \to \infty$. If $F(1) < 0$, the intermediate value theorem gives a root above 1. Since $F(1) = 1 - (1 - b)\big(\alpha + (\alpha - 1)W_\lambda(1)\big)$, $F(1) < 0$ exactly when the criterion fails. For $\lambda = 0$ and $\alpha > 1$, $W(z) \to +\infty$ as $z \to 1^+$.

For (3), with flat weights the homogeneous recurrence is $J_{n+1} - (1 + \alpha)J_n + J_{n-1} = 0$. Its two-sided bounded Green's function is $e^{-s|n-n_0|}/(2 \sinh s)$.

That the criterion is also sufficient is checked numerically (Table 6). □

6. Markov structure and loss

The framework has two Markov limits [3, Sec. 1]:

- the *forgetful* (Dirac) limit, in which nothing of the past enters;
- the *faithful* (light-cone) limit, in which every past contribution is retained distinctly. A closed system cannot reach it [3, Sec. 2].

The edge is neither of these.

Proposition 6.1 (The accumulator)**.** *As $\alpha \to 1^-$, $w_j \to 1$ for every fixed j. The memory becomes the running sum $\sum_{j \ge 1} J_{n-j}$, the discrete integral of the past. That is the accumulator of the fundamental theorem of calculus: it keeps the sum, not each contribution.*

In continuous lag, for $0 < \alpha < 1$,

$$\frac{\tau^{\alpha-1}}{\Gamma(\alpha)} = \int_0^\infty e^{-s\tau} \mu_\alpha(s)\, ds, \qquad \mu_\alpha(s) = \frac{\sin \pi\alpha}{\pi}\, s^{-\alpha}.$$

So a Markov description at $\alpha < 1$ needs a continuum of auxiliary modes $\dot{y}_s = -s y_s + J$. As $\alpha \to 1^-$ the weight μ_α concentrates at $s = 0$: on any interval $[0, S]$, the fraction of its mass in $[0, \delta]$ is $(\delta/S)^{1-\alpha} \to 1$. The continuum collapses onto the single mode $y_0 = \int^t J$, the accumulated state itself.

Proof. $\big(j^\alpha - (j-1)^\alpha\big)/\Gamma(\alpha+1) \to 1$. For the continuous part, use $\int_0^\infty s^{-\alpha} e^{-s\tau} ds = \Gamma(1-\alpha)\tau^{\alpha-1}$ and $\Gamma(\alpha)\Gamma(1-\alpha) = \pi/\sin \pi\alpha$. □

So the edge needs neither records nor extra state.

Theorem 6.2 (Normalizable memory has no undamped mode)**.** *Replace the power law by weights $w_j \ge 0$ with $\sum_j w_j = 1$. This includes every exponential kernel and the Dirac kernel $w_j = \delta_{j1}$. Then for every $\alpha \in (0, 1)$ and every $\hat{g} \in [0, 1]$, every root of F lies strictly inside the unit circle, so every mode decays.*

Proof. For $|z| \geq 1$:

$$|(1-\alpha)\hat{g}W(z)| \leq (1-\alpha)\sum_j w_j = 1-\alpha \leq |z|-\alpha \leq |z-\alpha|.$$

Equality throughout needs $z = 1$ and $\hat{g} = 1$, where $F(1) = 2(1-\alpha) \neq 0$. □

Remark 6.3 (Why the power law). The power law escapes Theorem 6.2 because its total weight is infinite, so $W(z)$ is unbounded as $z \to 1$. The vanishing factor $(1-\alpha)$ multiplies an unbounded memory and leaves the finite rotation $\sqrt{1-\alpha}$ per update. A normalizable memory damps every mode, at about $2(1-\alpha)$ per update when $(1-\alpha)$ times the mean lag is small (Table 7).

Remark 6.4 (Loss is contraction). In the framework, information is lost when a rank-2 object is contracted back to rank 1. That map is many-to-one: $a = \Omega u$ does not determine the pair (Ω, u) [3, Sec. 2].

- In the free edge the rank-2 factors $\Pi P \Phi$ are the identity, which is one-to-one. Nothing is erased, consistent with unitarity in the limit.
- Contraction becomes many-to-one in interaction, a rank-2 action on a rank-1 state, and in readout, where the closure $JJ^\dagger$ sends the 8 real components of J to 4. That is where the Landauer cost $k_B T \ln 2$ per erased bit is paid [7].
- The decay of Theorem 5.3 is a different thing. It is a phase effect of the fractional order, which vanishes at the edge.

7. Structure of the edge

7.1. **Causal structure: the edge is Galilean.** In the framework the Lorentz signature appears with the quadratic closure $JJ^\dagger$ at $\alpha = 2$, through the invariant $\det(JJ^\dagger) = |\det J|^2$ [1]. The same closure gives the Born readout through the trace, $\frac{1}{2}\mathrm{tr}(JJ^\dagger) = \sum_a |Z_a|^2$. That trace is a density, the time component of a current, not a Lorentz scalar. Below the closure there is no determinant, so there is no metric signature and no light cone.

At the $\alpha \to 1^-$ edge the causal structure is therefore time ordering alone: $J^-(x)$ reduces to the time-ordered past, with no bound on spatial reach. This is the Galilean structure of the Schrödinger equation, whose free propagator is nonzero at every distance for every $t > 0$.

- The lag-independent spatial profile of Definition 2.2 is admissible.
- A light cone imposed at the edge would import $\alpha = 2$ structure; that structure is treated in [4].

7.2. **The edge carries no interaction.**

Remark 7.1 (Single legs). In (1) the past enters only through transports from a point of the past to the present, each acting within one carrier. Hence:

(1) *No coupling between carriers.* For two carriers the update is a sum of single-carrier terms. It preserves the Schmidt coefficients between the carriers: it can carry entanglement that already exists, but cannot create any. The reason is the transport structure, not linearity. A linear rule with a two-carrier coupling would entangle.

(2) *A coupling would need the return leg.* A term transferring from A to B alone is one-way. Its adjoint is the return leg from B to A; for a unitary transport, $P_\gamma^\dagger = P_{\gamma^{-1}}$. A leg together with its return is order $1 + 1 = 2$: the reciprocal closure, which belongs to $\alpha = 2$ [4].

(3) *No holonomy from the carrier.* Holonomy needs a closed path [1]. What the edge can carry is a fixed background acting one way: a test particle that does not act back.

7.3. **Readout: one carrier, and composites.**

Proposition 7.2 (One carrier is two factors)**.** *Read $J \in M_2(\mathbb{C})$ as a vector $|J\rangle\!\rangle = \sum_{ij} J_{ij}|i\rangle|j\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^2$. The row index is the left factor, on which Lorentz transformations act ($J \to LJ$). The column index is the right factor, on which the internal phase acts ($J \to JU$). Then:*

1. *$JJ^\dagger = \mathrm{Tr}_{\mathrm{right}}|J\rangle\!\rangle\langle\!\langle J|$, the reduced state of the left factor;*
2. *J is a product state $\iff$ $\operatorname{rank} J = 1$ $\iff$ $\det J = 0$, which is a null closure* [3, Eq. (40)]*;*
3. *the eigenvalues of $JJ^\dagger$ are the Schmidt weights: their sum $\operatorname{tr} JJ^\dagger$ is the Born normalization, and their product $\det JJ^\dagger = |\det J|^2$ is the Lorentzian norm;*
4. *the concurrence of the normalized state is*

$$C = \frac{2|\det J|}{\operatorname{tr}(JJ^\dagger)} = \sqrt{1 - v^2/c^2} = \frac{1}{\gamma},$$

where $v/c = |\vec{x}|/x^0$ is the speed of the closure $JJ^\dagger = x^0 I_2 + \vec{x}\cdot\vec{\sigma}$.

Proof. For (1): $\left(\mathrm{Tr}_{\mathrm{right}}|J\rangle\!\rangle\langle\!\langle J|\right)_{ik} = \sum_j J_{ij}\bar{J}_{kj}$. For (2): a 2×2 matrix has rank one iff its determinant vanishes, and a rank-one matrix is a product $\psi\eta^\dagger$. Part (3) is the singular-value decomposition of J. For (4): the eigenvalues of $x^0 I_2 + \vec{x}\cdot\vec{\sigma}$ are $x^0 \pm |\vec{x}|$, and for a pure bipartite state $C = 2\sqrt{p_+ p_-}$ with $p_\pm = (x^0 \pm |\vec{x}|)/(2x^0)$. □

So "Born by trace" and "Lorentz by determinant" are the sum and the product of the Schmidt weights of a single carrier. A carrier at rest is maximally entangled between its factors, and a massless carrier is a product.

Proposition 7.3 (Lorentz action on the carrier's entanglement)**.** *Let $G \in SL(2,\mathbb{C})$ act on the left factor, $J \to GJ$. Then $Q \to GQG^\dagger$. This is the proper orthochronous Lorentz action on the Hermitian slice* [3, Eqs. (35)–(36)]*, and:*

1. *$|\det J|$, the invariant density, is unchanged, while $x^0 = \frac{1}{2}\operatorname{tr} Q$, the density in the frame, is not;*
2. *rotations, $G \in SU(2)$, are local unitaries and leave the concurrence unchanged; boosts, with G Hermitian and positive, are invertible local filtering operations and change it* [11]*;*
3. *writing the speed of the closure as $\beta = \tanh\eta$, with η its rapidity,*

$$C = \operatorname{sech}\eta, \qquad p_\pm = \frac{e^{\pm\eta}}{2\cosh\eta}, \qquad \frac{p_+}{p_-} = e^{2\eta},$$

so the Schmidt weights are proportional to the relativistic Doppler factors $e^{\pm\eta}$, and collinear boosts add rapidities;

4. *the pure states of the left factor, which are the null closures, form $\mathbb{CP}^1$, the celestial sphere, on which $SL(2,\mathbb{C})$ acts by Möbius transformations* [13].

Proof. $\det(GJ) = \det J$ when $\det G = 1$, and $Q \to GQG^\dagger$ preserves $\det Q = x_\mu x^\mu$. For $G \in SU(2)$ the trace is also preserved. For (3), use Proposition 7.2(4) with $\beta = \tanh\eta$. A boost $e^{\eta\, n\cdot\sigma/2}$ acting on a carrier at rest, $Q = \rho I$, gives $Q = \rho\, e^{\eta\, n\cdot\sigma}$, with $x^0 = \rho\cosh\eta$ and $|\vec{x}| = \rho\sinh\eta$. Part (4) is the spinor description of null directions. □

Remark 7.4 (What is known, and what is read)*.* The Lorentz structure of the qubit state space is known. The Bloch ball is the trace-one section of the future light cone of positive 2×2 matrices [12], and local filtering on a qubit acts as a Lorentz transformation [11].

What the carrier adds is a reading: the entanglement between a carrier's two factors is the time dilation of its current. This is a statement about the closure $JJ^\dagger$, an $\alpha = 2$ object. The Lorentz group here acts on the carrier's closure, not on the spacetime of the edge.

Proposition 7.5 (Readout under composition)**.** *Let $J \in M_N(\mathbb{C})$ have any rank, let $Q = JJ^\dagger$, and let P_m be projectors with $\sum_m P_m = I_N$. Then $p_m = \mathrm{Tr}(P_m Q)/\mathrm{Tr}\, Q$ is a probability distribution. For composite carriers $J_{AB} \in M_2(\mathbb{C}) \otimes M_2(\mathbb{C}) \cong M_4(\mathbb{C})$, and likewise for more carriers:*

1. *product carriers give product statistics;*
2. *marginals are consistent for every J_{AB};*
3. *the rank of J_{AB} and the entanglement between A and B are different things. For example, $J_{AB} = \Phi^+\eta^\dagger$ has rank one and gives a pure Q_{AB} with $\mathrm{Tr}_B Q_{AB} = I_2/2$.*

Proof. Positivity and $\sum_m P_m = I$ give a distribution. $(J_A \otimes J_B)(J_A \otimes J_B)^\dagger = J_A J_A^\dagger \otimes J_B J_B^\dagger$. Part (2) is the definition of the partial trace, and (3) is direct computation. □

Lemma 7.6 (Intrinsic normalization above M_2)**.** *The closure normalizes itself, $2J = JJ^\dagger$ giving a projector $P = J/2$ with $\mathrm{tr}\, P = 1$ [1], only if J has rank one.*

- *In $M_2(\mathbb{C})$, rank one is the condition $\det J = 0$.*
- *In $M_N(\mathbb{C})$ with $N \geq 3$, rank one is $\Lambda^2 J = 0$, the vanishing of all 2×2 minors. The condition $\det J = 0$ alone is not enough.*

Proof. $2J = JJ^\dagger$ forces $J = J^\dagger$ and $J^2 = 2J$, so $\mathrm{tr}\, P = \mathrm{rank}\, J$. For $N \geq 3$, $J = \mathrm{diag}(2, 2, 0, \ldots, 0)$ satisfies $2J = JJ^\dagger$ and $\det J = 0$, but $\mathrm{tr}\, P = 2$. □

Proposition 7.7 (Born needs one orientation)**.** *Let $J = \psi_a U + \psi_b V$ be an $M_2(\mathbb{C})$ carrier, with $Q = \sum_k JJ^\dagger$ and readout $p = \mathrm{tr}(PQ)/\mathrm{tr}\, Q$.*

1. *With a one-way past, $\mathrm{tr}\, Q$ decays at $2\big[(\pi/4)(1-\alpha) + \sqrt{1-\alpha}\,(\hbar/2m)\langle k^2\rangle\big]$ per radian. The readout drift vanishes as $\alpha \to 1^-$.*
2. *With a zero past, both rotation directions are present. Their cross terms oscillate at twice the rotation, $\mathrm{tr}\, Q$ swings by $O(1)$, and the readout drift does not vanish.*

Proof. For (1), apply Theorem 5.3 mode by mode. Part (2) and the numbers are in Table 5. □

This is the edge counterpart of the positive-frequency restriction at $\alpha = 2$ [4].

8. Numerical verification

All numbers come from `scripts/cal_edge_transducer.py` in the repository https://github.com/JewJitsu11B/Liorhybrid. It uses NumPy only and runs in about fifteen seconds. Methods are described in Appendix B.

Memory sum. Equation (2), with the Hurwitz zeta functions evaluated by Euler–Maclaurin, agrees with direct summation to 3×10^{-14} at $|z| > 1$. An independent arbitrary-precision evaluation agrees to 3×10^{-16}.

Wavepacket. The packet is $e^{-x^2/2+ix}$ on a periodic box of length 40 with 128 points. Its past is the carrier rotating one way over the whole infinite past, added exactly by Lemma 4.3(4). It is evolved by (1) to Schrödinger time $(\hbar/2m)t = 0.5$ and compared with free Schrödinger evolution after aligning the global phase.

$1-\alpha$	rotation/update	$\div\sqrt{1-\alpha}$	decay/update	decay per radian	$\div(1-\alpha)$
10^{-2}	0.09897219	0.989722	8.922×10^{-4}	9.015×10^{-3}	0.90146
10^{-3}	0.03157031	0.998341	2.652×10^{-5}	8.401×10^{-4}	0.84005
10^{-4}	0.00999775	0.999775	8.082×10^{-7}	8.084×10^{-5}	0.80843
10^{-5}	0.00316219	0.999972	2.512×10^{-8}	7.945×10^{-6}	0.79450
10^{-6}	0.00100000	0.999997	7.888×10^{-10}	7.889×10^{-7}	0.78885
10^{-7}	0.00031623	1.000000	2.488×10^{-11}	7.867×10^{-8}	0.78667

TABLE 1. Theorem 5.1 at $k = 0$. The rotation tends to $\sqrt{1-\alpha}$ per update, and the decay per radian divided by $1-\alpha$ tends to $\pi/4 = 0.7854$. Direct iteration of (1) at $\alpha = 0.999$ for 24000 updates gives rotation 0.031570 and decay 2.652×10^{-5}, the same as the root to every printed digit.

$\langle r^2\rangle$	k	measured	$\langle r^2\rangle/12$	ratio
0.06	0.10	0.005000	0.005000	1.0000
0.60	0.10	0.049988	0.050000	0.9998
0.60	0.20	0.049950	0.050000	0.9990
6.00	0.05	0.499688	0.500000	0.9994
6.00	0.20	0.495034	0.500000	0.9901

TABLE 2. Corollary 5.2 at $1-\alpha = 10^{-5}$: $\big(\varphi(0)-\varphi(k)\big)/\big(\varphi(0)k^2\big) \to \langle r^2\rangle/12 = \hbar/2m$. The departures are the k^4 terms.

$1-\alpha$	$k = 0.1$	$k = 0.3$	$k = 1$	$\sqrt{1-\alpha}$
10^{-2}	0.09187	0.09185	0.09163	0.10000
10^{-3}	0.03081	0.03080	0.03073	0.03162
10^{-4}	0.00992	0.00992	0.00990	0.01000
10^{-6}	0.00100	0.00100	0.00100	0.00100

TABLE 3. The diffusive admixture of Theorem 5.3: the extra decay of mode k divided by its kinetic phase, with $\langle r^2\rangle = 0.06$. The predicted value is $\frac{1}{2}\sqrt{\varepsilon}(1+\sqrt{\bar{g}}) - \frac{\pi}{4}\varepsilon$, for example 0.0921 at $\varepsilon = 10^{-2}$ and $k = 0.1$.

$1-\alpha$	$\langle r^2\rangle$	updates	$\|\mathrm{CAL}-\mathrm{Schr}\|$	floor (k^4)	counter-rotating
10^{-2}	0.06	1010	6.96×10^{-2}	1.14×10^{-2}	5.6×10^{-5}
10^{-3}	0.06	3168	2.63×10^{-2}	1.24×10^{-2}	7.2×10^{-7}
10^{-4}	0.24	2501	4.83×10^{-2}	4.81×10^{-2}	1.2×10^{-7}
10^{-4}	0.12	5001	2.57×10^{-2}	2.48×10^{-2}	4.3×10^{-8}
10^{-4}	0.06	10002	1.46×10^{-2}	1.26×10^{-2}	1.8×10^{-8}

TABLE 4. Corollary 5.2, Theorem 5.3 and Proposition 5.4. The floor is the distance between the exact modes and Schrödinger, which is the k^4 terms; it halves with $\langle r^2\rangle$. As $\alpha \to 1^-$ the error falls onto the floor. The norm loss at $1-\alpha = 10^{-4}$ is 0.9–1.5%: the uniform part plus the diffusive part $\sqrt{1-\alpha}\,\langle k^2\rangle(\hbar/2m)t$. With a two-cycle history instead, 70–97% of the norm goes into the counter-rotating mode.

	one-way past		zero past	
$1-\alpha$	$\operatorname{tr} Q$ drift per radian	readout drift	$\operatorname{tr} Q$ swing	readout drift
10^{-2}	2.1×10^{-2}	5.5×10^{-3}	3.5	3.7×10^{-2}
10^{-3}	2.8×10^{-3}	1.9×10^{-3}	2.2	3.6×10^{-2}
10^{-4}	5.5×10^{-4}	6.0×10^{-4}	2.0	3.7×10^{-2}

TABLE 5. Proposition 7.7, over 60 radians. With one orientation the closure and the readout become conserved as $\alpha\to1^-$. With both directions present they do not.

b	past never fades	$\lambda=0.1$	$\lambda=0.01$
0	+0.1010	+0.0647	+0.0964
0.05	+0.0775	+0.0293	+0.0715
0.2	+0.0356	−0.0456	+0.0270
0.5	+0.0103	−0.0874	+0.0005
0.9	+0.0012	−0.0987	−0.0088

TABLE 6. Theorem 5.5 at $\alpha=1.01$: growth per update (>0 runs away, <0 settles). The criterion gives $b>0.095$ for $\lambda=0.1$ and $b>0.501$ for $\lambda=0.01$. Without braking or fading the growth is 0.1010 against $\sqrt{\alpha-1}=0.1$, and at $\alpha=1.001$ it is 0.03167 against 0.03162. At $\alpha=1$ the carrier is frozen.

kernel ($\sum_j w_j=1$)	$1-\alpha$	largest $\lvert z\rvert$	slowest decay/update
exponential, width 10	10^{-2}	0.97677	2.35×10^{-2}
exponential, width 100	10^{-2}	0.99205	7.98×10^{-3}
summable power law $j^{-1.5}$	10^{-2}	0.98656	1.35×10^{-2}
Dirac, $w_j=\delta_{j1}$	10^{-2}	0.97979	2.04×10^{-2}
all four	10^{-4}	0.99980	2.0×10^{-4}

TABLE 7. Theorem 6.2: every root inside the unit circle, from the companion matrix of the recurrence with 400 lags.

Check	Result
$\lVert\mathrm{Tr}_{\mathrm{right}}\lvert J\rangle\!\rangle\langle\!\langle J\rvert - JJ^\dagger\rVert$, 200 random J	1.8×10^{-15}
$\lvert C-1/\gamma\rvert$, same J	1.2×10^{-15}
trace readout against $\lvert\Phi_m^\dagger\Psi\rvert^2/\Psi^\dagger\Psi$, $N\in\{2,4,16,1024\}$	3.7×10^{-16}
product carriers: $\lvert p(m,n)-p_A(m)p_B(n)\rvert$	1.1×10^{-16}
rank-4 J_{AB}: marginal mismatch	0
$J_{AB}=\Phi^+\eta^\dagger$: rank, $\mathrm{Tr}_B Q$	1, $I_2/2$
$\mathrm{diag}(2,2,0,0)$: $\det J$, largest 2×2 minor, $\operatorname{tr} P$	0, 4, 2
boost by $\eta=0.5,1,2$: C against $\operatorname{sech}\eta$	equal to 6 digits
boosts 0.7 then 0.9: C against $\operatorname{sech}1.6$	0.387978, 0.387978

TABLE 8. Propositions 7.2–7.5 and Lemma 7.6.

9. Remarks on scope

Remark 9.1 (What changes relative to [2] and to the earlier version). Seven things change.

- Posit 1 of [2] is replaced by the update rule, Axiom 3. The earlier field equation $T = 0$ is not used; T is the new state, and vanishes only when the present balances the past.
- The phase prefactor is removed. Its job is done by integrating the source's phase channel (Axiom 2) and by the orientation of the source's past (Axiom 4).
- The coupling constant is removed. In the free, flat sector the transport chain is the identity.
- The time unit and the double scaling are removed. Time is relational (Definition 2.4).
- Unitarity was exact for every $\alpha < 1$ in the earlier version, because the prefactor cancelled the phase of the memory. Without the prefactor it is exact only in the limit (Theorem 5.3).
- The earlier mass formula, which contained the coupling and the time unit, is replaced by $\hbar/2m = \langle r^2 \rangle/12$ in rest radians.
- The one-sidedness of the edge is now the feedback classification of Theorem 5.5.

Remark 9.2 (Relation to $\alpha = 2$). At $\alpha = 2$ the update rule is not the dynamics, since the present term alone doubles the current. There the balance reading applies: the present equals the light-cone accumulated past, which gives the Klein–Gordon equation and Feynman's path sum [4]. Exactly at 2 that law is conservative, and above 2 it runs away.

Remark 9.3 (The wave is the accumulated past). Proposition 5.4(4) shows that the whole one-way history is needed.

- Accumulation with real weights gives diffusion: the channel $e^{-H/\tau}$, as in Feynman–Kac.
- Accumulation with phase gives waves: the channel $e^{iS/\hbar}$, as in Feynman's path integral.
- The edge law of Theorem 5.3 is a mixture of the two, dominated by the phase.
- In Feynman's formulation, $\psi(x, t)$ is likewise the sum over the past parts of paths ending at (x, t) [5]. This is a parallel, not a derivation.

Remark 9.4 (A measurable departure). If physics sits at the edge, unitarity is exact. If it sits slightly below, Theorem 5.3 predicts a universal decay of $(\pi/4)(1-\alpha)$ per radian of rest rotation, plus a diffusion term of relative size $\sqrt{1-\alpha}$. Precision interference and clock comparisons would therefore bound $1 - \alpha$.

Remark 9.5 (Forward pointer: the equivalence condition). At the edge the inertial mass comes from the kernel width, and the rest rotation from the feedback rate. Nothing here ties them together. Requiring the width to be $\sqrt{6}$ Compton wavelengths is the statement that the inertial mass equals the rest energy divided by c^2, which a universal coupling of gravity to energy would require. This belongs to the $\alpha = 2$ program and is not claimed here.

10. Conclusion

The Causal Accumulation Law, iterated as a transducer, produces Schrödinger evolution at $\alpha \to 1^-$ with nothing added. The accumulated past enters with a negative sign below $\alpha = 1$ and pulls the current back. That feedback makes the current rotate by $\sqrt{1-\alpha}$ per update, and the rotation is the i of the Schrödinger equation. The kernel's spatial width slows the rotation at short wavelengths exactly as a kinetic term does, with $\hbar/2m = \langle r^2 \rangle/12$ in units of the rest rotation. So the mass is a property of the memory's spatial extent.

Unitarity is derived but holds only in the limit. Below the edge the fractional order leaks a uniform $(\pi/4)(1 - \alpha)$ per radian and adds a diffusive admixture of relative size $\sqrt{1-\alpha}$. Both vanish at the edge, which leaves the phase channel alone.

The direction of rotation, and with it the Born readout's stability, is inherited from the source's one-way past, and the whole past is needed. At $\alpha = 1$ the past decouples and the carrier freezes. Above it the feedback turns positive: a causal response runs away, and a bounded one anticipates its source, as for the Abraham–Lorentz–Dirac equation.

Normalizable memories cannot sustain the rotation. The power law can, because its total weight is infinite, and as $\alpha \to 1^-$ it becomes the accumulator of the fundamental theorem of calculus.

The edge is Galilean and free: interaction needs the return leg, which is order 2. Its readout is the closure of a carrier that is itself two factors, with concurrence $1/\gamma$.

Appendix A. Notation

α, $\varepsilon = 1 - \alpha$	fractional order and its distance to the edge
n, $t = \varphi_0 n$	update count and time in rest radians
w_j, $W(z)$	order-α memory weights and their generating function
g, $\hat{g}$, $\langle r^2 \rangle$	spatial profile, its Fourier transform, its second moment
$F(z)$	characteristic function of the update rule
φ	co-rotating root: rotation (real) and decay (imaginary) per update
$\hbar/2m = \langle r^2 \rangle/12$	kinetic coefficient in rest radians
b, λ	brake and fading rate (Theorem 5.5)
$\|J\rangle\!\rangle$	the carrier read as a vector in $\mathbb{C}^2 \otimes \mathbb{C}^2$
$Q = JJ^\dagger$, P_m	positive closure; outcome projectors
C, γ, η	concurrence; Lorentz factor and rapidity of the closure

Appendix B. Numerical methods

Roots. Newton iteration on $F(e^{i\varphi})$, with W evaluated by (2). The two Hurwitz zeta functions are evaluated by Euler–Maclaurin with 30 terms and seven Bernoulli corrections, and checked against an independent arbitrary-precision library.

Iteration. Equation (1) is iterated with the full history, an $O(N^2)$ direct sum. A one-way past is added exactly through Lemma 4.3(4), using cumulative partial sums of $w_j e^{-i\varphi_h j}$.

Packets and closure.

- The packet runs keep the Fourier modes with amplitude above 10^{-9} of the maximum.
- The counter-rotating fraction is fitted over the last two cycles.
- The closure test uses $J = \psi_a U + \psi_b V$, with random $U, V \in M_2(\mathbb{C})$, packets centred at $k_0 = 1$ and $k_0 = -1.5$, and a random rank-one projector P.

Feedback and normalizable kernels.

- Growth rates are least-squares slopes of $\log|J_n|$ over the second half of 6000 updates.
- The normalizable-kernel roots are eigenvalues of the companion matrix of the recurrence truncated at 400 lags.

Readout. Partial traces by index contraction; outcome bases from QR factorization of random complex matrices; boosts $e^{\eta\, n\cdot\sigma/2}$ along a random axis.

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